

LocalMesh

Private Network Service Platform

Phase 1 architecture

Harsith (DNS + UTM host) · Manohar (nginx edge)

app.localmesh.test 10.7.12.104 10.7.2.38

1 Network topology

Two physical Macs share one Wi-Fi LAN. Ubuntu backends are **UTM guests inside Your Mac**, not extra Wi-Fi hosts. Manohar reaches them only as 10.7.12.104:3001 and :3002. Guests (10.0.2.x) are not pingable from the LAN. Clients never type a backend address. HTTPS always goes **client** → **Manohar :8443** first. The dashed arrow is a DNS A-record only.

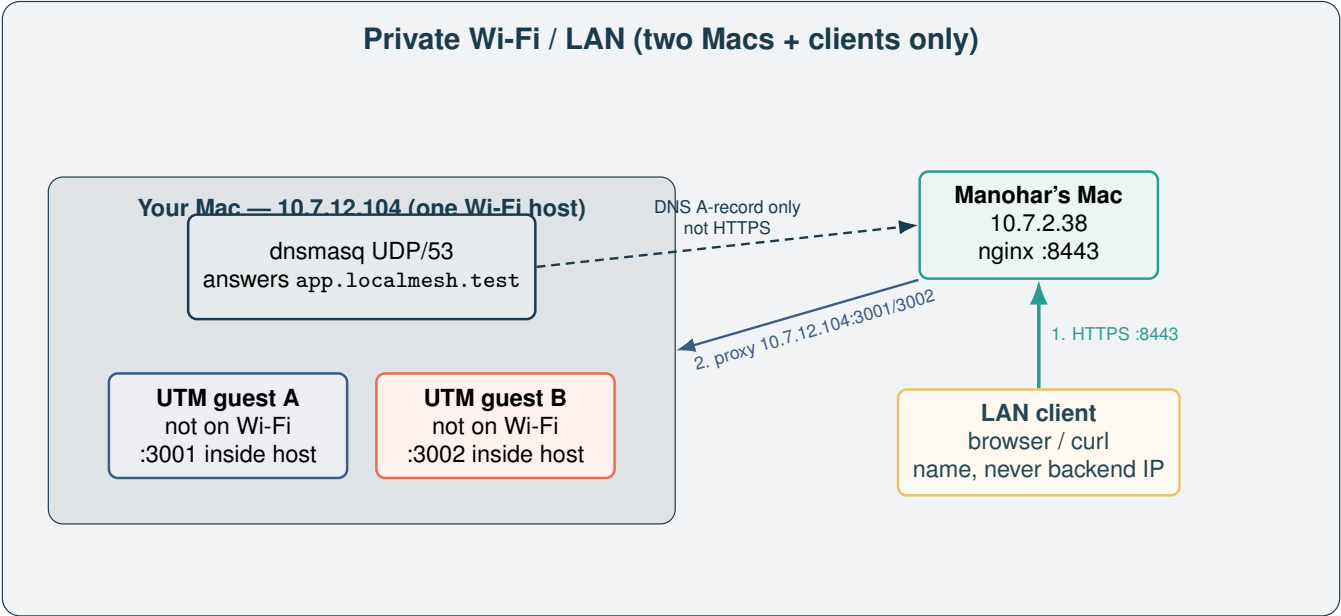


Figure 1: LAN topology. Dashed = DNS A-record to 10.7.2.38. Solid teal = HTTPS to nginx. Backends sit inside Your Mac.

2 One request

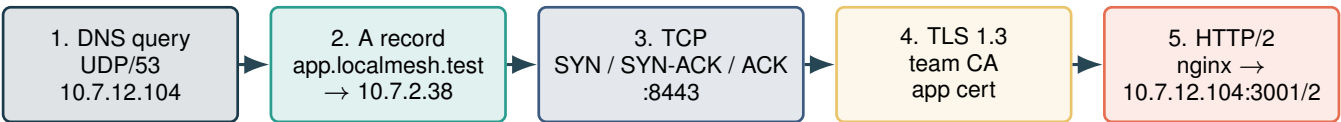


Figure 2: DNS → TCP → TLS → HTTP/2 → Backend A or B.

3 Inventory

Brief	Machine	LAN identity	Role
Mac 1	Your Mac	10.7.12.104	dnsmasq :53, UTM host
Mac 2	Manohar's Mac	10.7.2.38	nginx :8080/:8443, TLS, LB
Mac 3	UTM VM (3001)	10.7.12.104:3001	Backend A
Mac 4	LocalMesh-1	10.7.12.104:3002	Backend B

Settings from ~/cn-team.env: TEAM=localmesh, YOUR_MAC_IP=10.7.12.104, MANOHAR_MAC_IP=10.7.2.38, COLLEGE_DNS=8.8.8.8.

4 UTM publication

Shared Network has no Port Forward table. Emulated VLAN publishes guests on the host Wi-Fi IP.

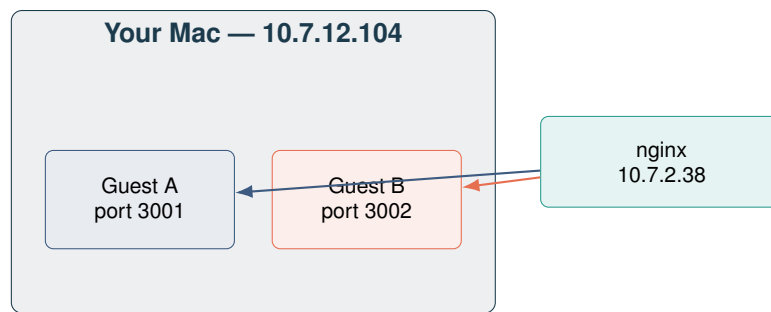


Figure 3: Host ports 3001 and 3002 are the only backend addresses nginx uses.

5 Layers

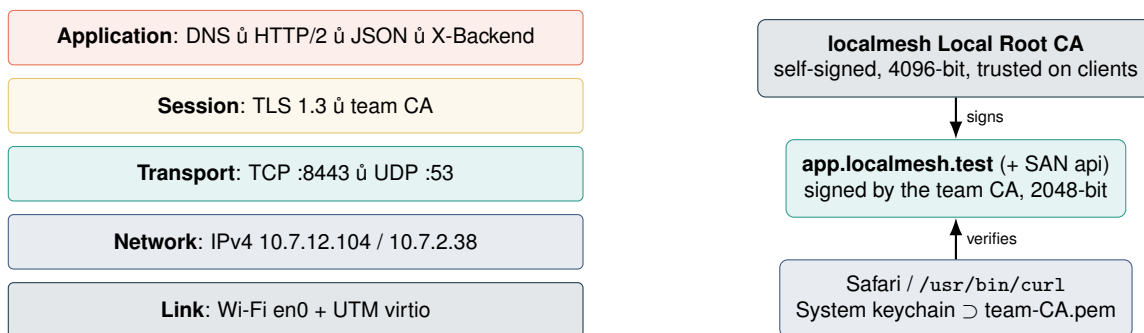


Figure 4: Left: OSI / TCP-IP mapping. Right: private CA trust.

6 Why this satisfies the brief

- Four *roles* exist. Hardware may be combined; the backends remain two processes on two ports.
- Emulated VLAN is hypervisor hosting, not a NAT protocol we implemented.
- Ports 8080/8443 are permitted. The zone is `.test`, never `.local`.
- Evidence: `dig to 10.7.2.38 via 10.7.12.104#53`; TLSv1.3; issuer `localmesh Local Root CA`; X-Backend A/B.